

Mathematics for Mechatronics Engineering

Calculus and Differential Equations

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- **Calculus Session 1:** Derivatives and Optimization
- **Calculus Session 2:** Integrals and Engineering Models
- **Differential Equations Session 1:** First-Order Systems
- **Differential Equations Session 2:** Second-Order Systems

Goal: Use mathematics to model and understand engineering systems.

Why Engineers Use Calculus

Calculus studies change and accumulation.

Engineering Applications

- Robot motion
- Velocity and acceleration
- Optimization
- Sensor measurements
- Dynamic systems

Calculus helps engineers predict system behavior.

Derivatives as Rates of Change

Derivative:

$$\frac{dx}{dt}$$

measures how fast a quantity changes.

Example:

$$x(t) = 3t^2 + 2t$$

Velocity:

$$v(t) = \frac{dx}{dt} = 6t + 2$$

Acceleration:

$$a(t) = \frac{d^2x}{dt^2} = 6$$

Physical Meaning

- Position $\rightarrow$ derivative gives velocity
- Velocity $\rightarrow$ derivative gives acceleration
- Temperature $\rightarrow$ derivative gives heating rate

Derivatives describe system dynamics.

Partial Derivatives

Many engineering systems depend on multiple variables.

$$f(x, y) = x^2 + 3xy + y^2$$

Partial derivative with respect to x :

$$\frac{\partial f}{\partial x} = 2x + 3y$$

Partial derivative with respect to y :

$$\frac{\partial f}{\partial y} = 3x + 2y$$

Interpretation

Partial derivatives measure sensitivity to one variable.

Gradient

The gradient is:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Example:

$$f(x, y) = x^2 + y^2$$

$$\nabla f = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

Meaning

The gradient points in the direction of steepest increase.

Engineers often minimize or maximize functions.

Example:

$$f(x) = x^2 - 4x + 1$$

Find minimum:

$$\frac{df}{dx} = 2x - 4$$

Set derivative equal to zero:

$$2x - 4 = 0$$

$$x = 2$$

Examples

- Minimize energy consumption
- Optimize robot trajectories
- Tune controller parameters
- Machine learning optimization

Linearization and Taylor Approximation

Approximate nonlinear functions near an operating point.

$$f(x) \approx f(a) + f'(a)(x - a)$$

Example:

$$\sin x \approx x$$

near $x = 0$.

Why Engineers Care

Many nonlinear systems are approximated as linear systems.

Taylor approximation in MATLAB:

```
syms x  
f = sin(x);  
taylor(f)
```

Output:

x

Practice Exercise

Given:

$$x(t) = 2t^3 + 5t$$

Find:

- ① Velocity
- ② Acceleration

Also compute:

$$\frac{\partial}{\partial x}(x^2y + 3y^2)$$

Physical Meaning of Integrals

Integrals measure accumulation.

$$\int f(x) dx$$

Engineering Examples

- Distance from velocity
- Energy consumption
- Total electrical charge
- Heat accumulation

Example: Distance from Velocity

Suppose:

$$v(t) = 2t$$

Distance traveled:

$$x(t) = \int 2t \, dt = t^2$$

Interpretation

Integration accumulates small changes over time.

Definite Integrals

Definite integral:

$$\int_a^b f(x) dx$$

computes total accumulated quantity between two points.

Example:

$$\begin{aligned} \int_0^2 2x dx \\ = x^2 \Big|_0^2 = 4 \end{aligned}$$

Real engineering data is often discrete.

Common Numerical Methods

- Trapezoidal rule
- Simpson's rule

Example:

$$\int_a^b f(x) dx \approx \frac{h}{2}(f_0 + f_1)$$

Numerical methods are heavily used in simulations.

Multivariable Integration

Double integrals accumulate quantities over areas.

$$\iint_A f(x, y) dA$$

Engineering Applications

- Mass distribution
- Pressure over surfaces
- Heat transfer

Differential Equations as Models

Many engineering systems are modeled using differential equations.
Example:

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = F(t)$$

- m : mass
- b : damping
- k : spring stiffness

Electrical circuits also produce differential equations.

$$RC \frac{dv}{dt} + v = V_s$$

Interpretation

The voltage changes gradually rather than instantly.

Numerical integration in MATLAB:

```
t = 0:0.1:10;  
v = 2*t;  
trapz(t,v)
```

Practice Exercise

Compute:

$$\int_0^3 (2x + 1) dx$$

Then explain the physical meaning if:

$$f(x) = \text{velocity}$$

What is a Differential Equation?

A differential equation contains derivatives.

Example:

$$\frac{dx}{dt} = 3x$$

Interpretation

The rate of change depends on the current value.

Differential equations describe dynamic systems.

Examples

- RC circuits
- Robot motion
- Thermal systems
- Motor dynamics
- Vibrations

Solving Simple Differential Equations

Example:

$$\frac{dx}{dt} = 3x$$

Separate variables:

$$\frac{dx}{x} = 3dt$$

Integrate:

$$\ln x = 3t + C$$

Solution:

$$x = Ce^{3t}$$

First-Order Linear Differential Equations

General form:

$$\frac{dy}{dt} + ay = f(t)$$

Engineering Example

RC circuit:

$$RC \frac{dv}{dt} + v = V_s$$

Time Constants

Example solution:

$$x(t) = e^{-t/\tau}$$

- Large τ : slow response
- Small τ : fast response

Engineering Meaning

Time constants determine system speed.

Stable system:

$$e^{-t} \rightarrow 0$$

Unstable system:

$$e^t \rightarrow \infty$$

Important

Stability is critical in control systems and robotics.

Plot exponential decay:

```
t = 0:0.1:10;  
x = exp(-t);  
plot(t,x)
```

Practice Exercise

Solve:

$$\frac{dx}{dt} = 2x$$

Then determine whether the system is:

- Stable
- Unstable

Second-Order Systems

Many mechatronic systems are second-order systems.
General form:

$$a \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + cx = f(t)$$

Mass-Spring-Damper System

Example:

$$m\ddot{x} + b\dot{x} + kx = 0$$

where:

- m : mass
- b : damping
- k : stiffness

Characteristic Equation

Assume solution:

$$x = e^{\lambda t}$$

Substitute into differential equation:

$$a\lambda^2 + b\lambda + c = 0$$

Solve for roots λ .

Types of Responses

Overdamped

Two real negative roots.

Critically Damped

Fastest non-oscillatory response.

Underdamped

Complex roots produce oscillation.

Natural Frequency

Natural frequency:

$$\omega_n = \sqrt{\frac{k}{m}}$$

Engineering Meaning

Systems naturally oscillate at specific frequencies.

Resonance occurs when forcing frequency matches natural frequency.

Examples

- Vibrating structures
- Motor vibrations
- Mechanical resonance

Damped oscillation:

```
t = 0:0.01:10;  
x = exp(-0.2*t).*cos(2*t);  
plot(t,x)
```

Practice Exercise

Given:

$$\ddot{x} + 4\dot{x} + 5x = 0$$

Find:

- 1 Characteristic equation
- 2 Roots
- 3 System type

Summary

- Calculus models change and accumulation
- Optimization is essential in engineering design
- Differential equations describe dynamic systems
- Stability and oscillation are key engineering concepts